



Influence of Room Acoustics on Objective Voice Assessment Methods in the Context of Speech and Language Therapy

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Abstract

Voice and speech disorders can be assessed through subjective perceptual measures or objective indices, such as the Acoustic Voice Quality Index (*AVQI*) or the Acoustic Breathiness Index (*ABI*). These objective measures reduce diagnostic variability by eliminating subjective evaluations. However, the impact of room acoustics on their robustness for therapy monitoring remains unclear. In this contribution, reverberation times, impulse responses, and background noise were recorded and analyzed in conjunction with speech samples using statistical models to evaluate their influence. Results indicate that room acoustics significantly affect *AVQI* and *ABI*, particularly for non-pathological voices and with distant microphones. For reliable therapeutic use, standardized measurement environments or robust analysis methods are essential. Optimized recording conditions with source-proximate microphones enhance accuracy, advancing objective voice quality assessments and evidence-based speech therapy.

Index Terms: Voice Disorders, Objective voice quality analysis, Room Acoustics

1. Introduction

In the diagnostic assessment and monitoring of voice and speech disorders in speech and language therapy (SLT), both subjective and objective evaluation methods are employed. Subjective methods, such as perceptual assessment of voice quality (e.g., through audiovisual perception, videostroboscopy, or patient self-assessments), are often complemented by objective measurements that capture acoustic parameters and airflow during phonation [1].

Standardized procedures such as the GRBAS scale (grade of hoarseness, roughness, breathiness, asthenicity and strain) and CAPE-V (Consensus Auditory-Perceptual Evaluation of Voice) [2] enable a structured evaluation, yet they exhibit limited interrater reliability [3]. As an objective alternative, indices such as the Dysphonia Severity Index (DSI) [4], the Acoustic Voice Quality Index (*AVQI*) [5], and the Acoustic Breathiness Index (*ABI*) [6] have been developed, representing voice quality as a single, comparable numerical value.

A significant challenge of these objective methods is achieving robust and comparable results in the acoustically diverse conditions commonly encountered in clinical and ambulant therapy settings, which are often dictated by practical necessities rather than optimal acoustic design. Variations in room configurations and ambient noise (e.g., from adjacent rooms or street traffic) increase the variability of the measurement conditions, complicating the application of thresholds to distinguish between pathological and non-pathological voices and to objec-

tively assess therapy outcomes.

Uloza et al. [7] demonstrated that the impact of different microphones and artificially generated environmental noise on the *AVQI* is minimal and statistically not significant. However, it is important to note that the added noise was at a very low level, and the recordings were conducted in a soundproofed environment. This is still close to the original recording conditions during the development of the original *AVQI* and *ABI* [8][6]. This study for the first time determines extensive acoustic room parameters in SLT rooms and evaluates their impact on objective voice quality measures.

The present paper is organized as follows: Section 2 introduces the measurement and analysis procedures, while Section 3 presents the results along with an extensive discussion. Finally, Section 4 provides the conclusions drawn from the paper's findings.

2. Methodology

Acoustic measurements were conducted using a *G.R.A.S. Knowles Electronics Manikin for Acoustic Research (KEMAR) Head and Torso Type 45BM*, dressed in a T-shirt, in combination with a *G.R.A.S. 44AB Mouth Simulator*. This ensures reproducible impulse responses (*IR*) that realistically simulate speech therapy settings.

The mouth simulator possesses frequency-dependent transmission characteristics that needed to be equalized and calibrated for this study. Following equalization, the resulting transfer function exhibited a deviation of less than 2 dB across all one-third octave frequencies within the range of $f_{min} = 50\text{Hz}$ to $f_{max} = 10\text{kHz}$.

2.1. Impulse Response Measurements in Therapy Rooms

The *IR* were measured in eight regionally and randomly selected institutions with a total of 35 rooms (32 speech-language therapy rooms, 3 audiology test rooms), placing the KEMAR in the designated patient seating position. Excitation was performed using a sequence of ten logarithmically swept sine signals in the relevant frequency range of $f_{min} = 50\text{Hz}$ to $f_{max} = 10\text{kHz}$, weighted with the equalization *IR* of the mouth simulator.

To analyze quality measures based on microphone type and position, sweep signals were recorded simultaneously using the following microphones:

- A *DPA 4006-TL Studio Microphone with DD0251 free-field grid* used as level calibration reference and for reverberation time measuring, placed on the treatment table in front of the KEMAR.

- A *Motorola moto g84 5G (XT2347-2)* smartphone, positioned closed to the reference microphone on the table.
- A *RODE WIRELESS PRO* lavalier microphone, attached to the collar of the T-shirt.

All microphones were aligned toward the KEMAR. The lavalier microphone was positioned approximately 20 cm from the mouth opening, while the other microphones were placed on the treatment table at realistic distances corresponding to typical speech quality assessments in therapy (approximately 0.4 to 0.6 m selected to reflect real-life conditions).

2.2. Reverberation Times and Background Noise

Reverberation times were determined for 23 third-octave bands with center frequencies ranging from $f_{m_1} = 64\text{Hz}$ to $f_{m_{23}} = 10\text{kHz}$ using Schroeder integration of the squared absolute signal [9], allowing for comparison with the recommendations of *DIN 18041*.

To assess background noise in the SLT rooms, recordings were made using all microphones at least for 60 s without any activity in the room. This was done during regular clinical operations to obtain a representative sample under real-world conditions. The recorded levels of the differently calibrated microphones were then harmonized using broadband adjustments and subsequently calibrated in dB_{SPL} .

As a reference for microphone level calibration, white noise was played and measured with the *DPA 4006-TL Studio Microphone* using a *WEVOSYS lingWAVES SPL meter II Sound Level Meter* in dB_{SPL} with C-weighting and “Slow” analysis settings. The sound source was a *Norsonic Dodecahedron Speaker Nor229*, driven by a *Norsonic Power Amplifier 260* positioned behind the KEMAR whenever possible.

2.3. Speech Signal Leveling and Data Analysis

The levels of the speech signals used were adjusted across different microphone positions by determining the mean speech level using a voice-activity detector, based on a power threshold of -30dB relative to the maximum speech signal power, and normalizing it to 80dB_{SPL} for the lavalier microphone signal. At a 20 cm distance, this corresponds to an average speech sound level of approximately 65dB_{SPL} at one meter. The levels of the other microphones were adjusted according to their measured transfer functions.

The impact of room acoustics on the objective speech quality measures *AVQI* and *ABI*, along with their underlying parameters, was analyzed using the *Praat* script *praat_analysis.praat*, integrated into the *VOXplot* software [10]. Measurements were performed for:

- Dry speech signals,
- Speech signals convolved with *IRs*
- Speech signals mixed with randomly selected background noise from the measured noise floor.

This method enables realistic simulation by computing speech quality measures directly within the captured clinical environments while allowing independent analysis of room acoustics and noise effects.

The influence of room acoustics on the voice quality parameter X is analyzed via the difference:

$$\Delta_X = X_{\text{room}} - X_{\text{dry}},$$

where X_{dry} is the measure without and X_{room} the measure with room effects. Positive Δ values indicate deterioration. For this

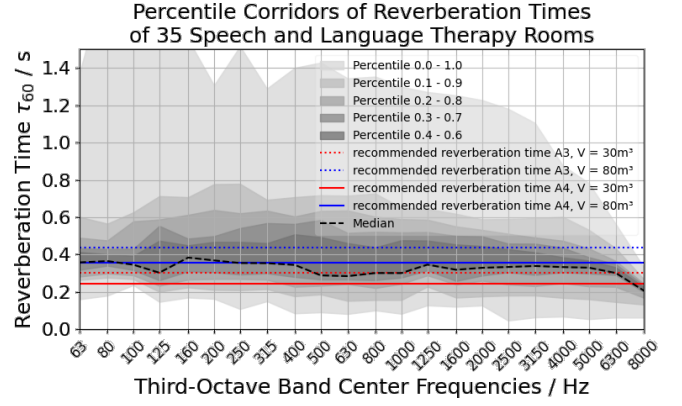


Figure 1: *Percentile Corridors of reverberation times as a function of third-octave band center frequencies for 35 speech and language therapy rooms and recommended reverberation times for usage types A3 and A4 according to DIN 18041.*

analysis, $N = 1644$ voice samples from the Saarbrücken Voice Database (SVD) [11] were used.

The analysis of acoustic influences on speech quality measures and parameters was conducted using mixed-effects linear models (MixedLM) with restricted maximum likelihood (REML) estimation [12]. Model performance was evaluated based on the normalized Root Mean Squared Error ($\|RMSE\|$) and the coefficient of determination (R^2) [13]. Various model variants, incorporating both linear and nonlinear terms (e.g., $\log(X)$, x^2 , x^3), were explored. After model selection, predictions were made for a test data subset to assess accuracy and determine the optimal model.

3. Results and Discussion

The presentation of the results is divided into two sections: background noise analysis and the impact of room acoustics.

3.1. Background Noise

The noise floor measurement revealed that in 27 of 35 rooms the background noise level is $\leq 40\text{dB}(A)$, in five rooms it is $\leq 45\text{dB}(A)$, and only one room exceeds $55\text{dB}(A)$. All values are higher compared to the simulated noises in [7].

The analysis of the noise floor showed only minor deviations in *AVQI* for both microphones: For the lavalier microphone the mean value deviates $\mu_{\Delta, \text{Noise}} = 0.10$ units (interquartile range $IQR = 0.10$), and for the smartphone, $\mu_{\Delta, \text{Noise}} = 0.07$ ($IQR = 0.08$). Similar values were observed for *ABI*: Lavalier: $\mu_{\Delta, \text{Noise}} = 0.07$ ($IQR = 0.15$), Smartphone: $\mu_{\Delta, \text{Noise}} = 0.03$ ($IQR = 0.12$). Outliers (approximately ± 1 unit) occasionally occurred; notably, Institution 5, Room 1, exhibited a significantly higher mean and increased *IQR*. This very large SLT room is primarily used for gymnastics and only serves as a backup SLT room. These small deviations underlines the results of [7] that the impact of noise at typical background level is only of secondary importance for the usage of *AVQI* and *ABI* as diagnostic tools.

3.2. Room Acoustics

Figure 1 shows the relative frequencies of reverberation times in 100 ms intervals, broken down by selected third-octave band center frequencies.

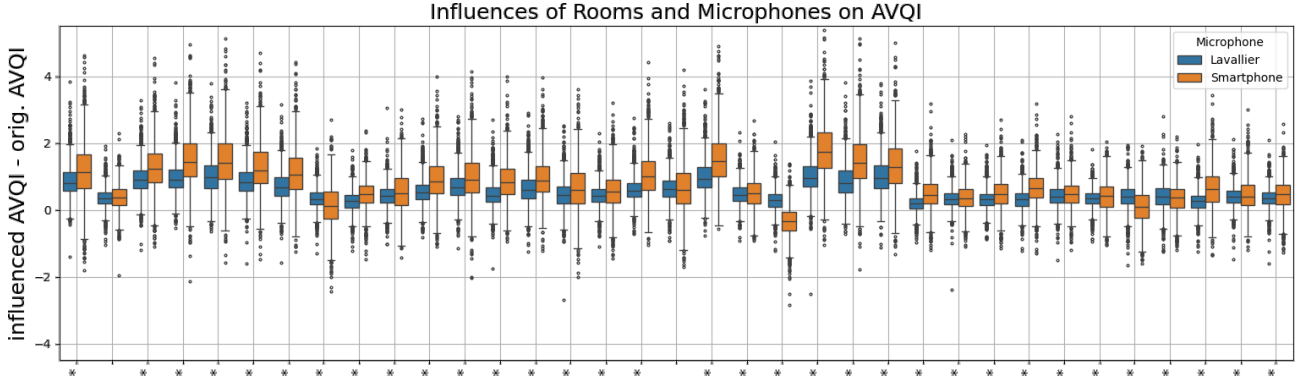


Figure 2: Distributions of the influence of room acoustics on the objective vocal quality measures AVQI for voice samples from the Saarbrücken Voice Database, determined by the difference in vocal quality measures with and without room influence, depending on the respective room and different microphones. The differences between the microphones in each room were tested for statistical significance using a t-test ($p < 0.05$) and marked with * if distributions significantly differed.

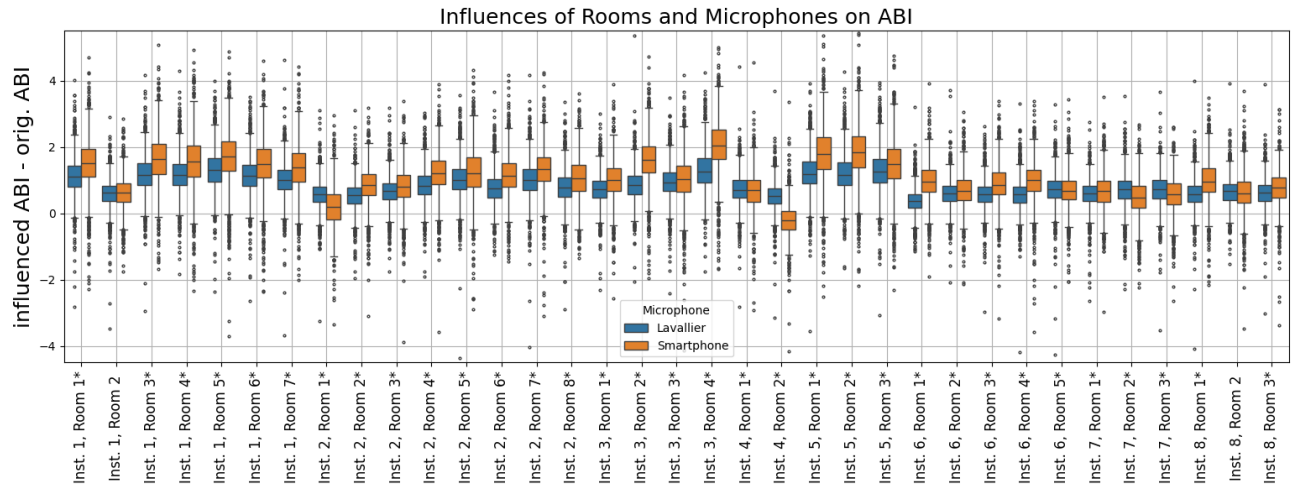


Figure 3: Distributions of the influence of room acoustics on the objective vocal quality measures ABI for voice samples from the Saarbrücken Voice Database, determined by the difference in vocal quality measures with and without room influence, depending on the respective room and different microphones. The differences between the microphones in each room were tested for statistical significance using a t-test ($p < 0.05$) and marked with * if distributions significantly differed.

Despite the specific requirements for treating voice and speech disorders, there are no binding acoustic standards for SLT rooms in Germany. Existing regulations only specify minimum room sizes: at least 20 m^2 for a primary treatment room and 12 m^2 for additional rooms [14].

The DIN 18041:2016-03 [15] provides guidelines for optimal room acoustics. SLT rooms fall under Category A – “Rooms for Communication” – and should ideally meet usage type A3 (“Teaching/Communication”) or, preferably, A4 (“Inclusive Teaching/Communication”), as SLT therapists often work with patients with hearing impairments and/or special needs. The standard defines recommended reverberation times for each usage type with a frequency-dependent tolerance of $\pm 20\%$ within the 250 to 2000 Hz range. We assume, comparable guidelines are available in other countries, perhaps less officially defined.

With an average room height of $\approx 2.5 \text{ m}$, a typical room volume ranges from $V = 30$ to 80 m^3 . For comparison purposes, Figure 1 includes the recommended reverberation time thresholds for these two boundary values.

Only eight of the measured rooms meet or fall below the recommended reverberation times for A4, while approximately half adhere to A3 standards. This highlights that room acoustics may not be the primary concern when planning the interior of SLT rooms.

Figures 2 and 3 show the AVQI and ABI, respectively, for all 35 rooms. For each room, two boxplots display the Δ_X values for all sentences in the SVD, corresponding to the two tested microphone positions. The boxplots display the mean, IQR, and individual outliers. Additionally, asterisks below the boxplots indicate whether the results for the two microphone positions are statistically different. The t-test results were computed at a significance level of 0.05.

These Figures illustrate that, with one exception, voice quality measures deteriorate by 0 to 2 units on average in all rooms, with a maximum IQR of about 1 unit and outliers up to ± 6 units. Moreover, the impact of room acoustics is lower for the lavalier microphone than for the smartphone, as confirmed by a t-test (significance level 0.05). Furthermore, the higher distance to the mouth also increases the IQR and variance for all

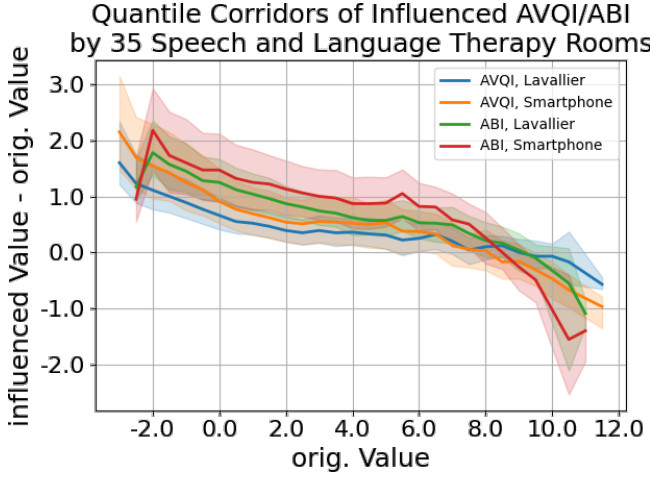


Figure 4: Interquartile range and medians of room acoustics-influenced voice quality measures (AVQI/ABI) as a function of the unaffected voice quality measure for different microphones in 35 speech therapy rooms.

Table 1: Model fitting results for the objective vocal quality measures AVQI and ABI, along with their individual parameters, including the normalized RMSE, the coefficient of determination R^2 , and the results of the Kolmogorov-Smirnov test (test statistic W - and p -value)

Parameter X	$\ RMSE\ $	R^2	W	p
AVQI	0.636	0.596	0.279	< 0.01
ABI	0.669	0.553	0.249	< 0.01

measurements.

The striking similarity between the two illustrations highlights the close relationship between AVQI and ABI, as both indices depend on similar low-level features for their final calculations. The dependency of the acoustically influenced Δ on the dry voice quality measures AVQI and ABI, as well as for different microphones, is shown in Figure 4. It can be observed that Δ varies with the unaffected voice quality measures: for lower AVQI/ABI values (less pathological voices), the mean deviations are approximately 2 units higher than for more pathological voices. In some cases, very high dry values even result in an improved quality assessment solely due to room effects.

Based on these observations, a mixed-effects linear model (MixedLM) using the `statsmodels` library (v0.14.4) in Python was selected and evaluated, accounting for the fixed effects of categorical room C (Room) dependency, the influence of the dry voice quality measure X , and their interaction, as well as the random effect of subjects:

$$\Delta = X * C(\text{Room}) + (1|\text{subject}). \quad (1)$$

Exploratory analysis revealed that the voice quality measures are non-normally distributed due to skewness and outliers. Examination of the residuals (the differences between observed and predicted values) showed that they are symmetric but narrower and more concentrated around the mean than a normal distribution (Kolmogorov-Smirnov test, p -value < 0.01). These results suggest that, despite slight deviations, the model is robust and permits reliable statistical inferences.

The following results refer to the smartphone microphone, as the results for the lavalier microphone—while comparable—are less pronounced. Table 1 summarizes key model fit metrics ($\|RMSE\|$, R^2 , and the results of the Kolmogorov-Smirnov test) for the different voice quality parameters. The models exhibit a similar prediction error ($\|RMSE\|$) of approximately 0.6 units for both overall measures. A moderate yet acceptable model fit of $R^2 > 0.55$ suggests that the model captures a significant portion of the variability in voice quality assessments, with the remaining variance attributed to unaccounted factors or random influences.

The effect of the dry voice quality measure is significant ($p < 0.01$) and negative for both AVQI (−0.245) and ABI (−0.233). This confirms that deviations are larger for lower dry values and decrease with increasing values. Room effects range from −2.294 to 0.620 and are significant for almost all rooms ($p < 0.01$) – except for *Institut 1 Room 3* and *Room 6* (AVQI) and *Institut 3 Room 3* and *Institut 5 Room 3* (ABI) – indicating that some rooms positively, and others negatively influence Δ . Numerous interactions are also significant ($p < 0.01$), demonstrating that the effects of the acoustic environment vary with the dry values. The group level variance is approximately 0.15 for both measures, indicating moderate individual variability.

4. Conclusions

While background noise levels in momentary recordings are generally acceptable, excessive reverberation is particularly evident in SLT rooms treating children with hearing impairments or auditory processing disorders (APD) [16] as well as hearing impaired elderly patients.

Our results show that most of the examined rooms are acoustically suboptimal, with recommended reverberation times frequently exceeded.

The study shows that the voice quality measures AVQI and ABI are significantly influenced by room acoustics, the degree of voice impairment, and the microphone position. On average, the measurements shift by 1 to 2 units (on a scale from 0 to 10), with observed deviations ranging from −2 to +4 units. Notably, voices that are unremarkable (non-pathological) tend to be more affected, which can lead to a classification in the pathological range.

Furthermore, the impact of room acoustics also depends on the microphone position: the closer the microphone is to the sound source (e.g., a lavalier microphone), the smaller the deviation from the dry signal and the lower the dispersion of the measurements. Thus, in addition to optimizing room acoustics, the use of source-proximate microphones – ideally headset microphones – should be considered when assessing voice quality.

In summary, comparing objective voice quality measures between different institutions or even between rooms within the same institution is problematic due to room acoustic differences. However, AVQI and ABI can be used for monitoring therapy progress, provided that measurements are consistently conducted under identical room and recording conditions. For automated and technology-assisted analysis methods, it is essential either to optimize the acoustic properties of SLT rooms or to develop measurement techniques that are more robust to room influences. Additionally, therapists should be informed about the potential effects of room acoustics to ultimately enhance the reliability and accuracy of the results.

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6. References

- [1] P. H. Dejonckere, P. Bradley, P. Clemente, G. Cornut, L. Crevier-Buchman, G. Friedrich, P. Van De Heyning, M. Remacle, V. Woisard, and Committee on Phoniatrics of the European Laryngological Society (ELS), "A basic protocol for functional assessment of voice pathology, especially for investigating the efficacy of (phonosurgical) treatments and evaluating new assessment techniques. Guideline elaborated by the Committee on Phoniatrics of the European Laryngological Society (ELS)," *European archives of oto-rhino-laryngology: official journal of the European Federation of Oto-Rhino-Laryngological Societies (EUFOS): affiliated with the German Society for Oto-Rhino-Laryngology - Head and Neck Surgery*, vol. 258, no. 2, pp. 77–82, Feb. 2001.
- [2] G. B. Kempster, B. R. Gerratt, K. Verdolini Abbott, J. Barkmeier-Kraemer, and R. E. Hillman, "Consensus Auditory-Perceptual Evaluation of Voice: Development of a Standardized Clinical Protocol," *American Journal of Speech-Language Pathology*, vol. 18, no. 2, pp. 124–132, May 2009, publisher: American Speech-Language-Hearing Association. [Online]. Available: <https://pubs.asha.org/doi/10.1044/1058-0360%282008/08-0017%29>
- [3] J. Kreiman, B. R. Gerratt, G. B. Kempster, A. Erman, and G. S. Berke, "Perceptual evaluation of voice quality: review, tutorial, and a framework for future research," *Journal of Speech and Hearing Research*, vol. 36, no. 1, pp. 21–40, Feb. 1993.
- [4] F. L. Wuyts, M. S. De Bodt, G. Molenberghs, M. Remacle, L. Heylen, B. Millet, K. Van Lierde, J. Raes, and P. H. Van de Heyning, "The dysphonia severity index: an objective measure of vocal quality based on a multiparameter approach," *Journal of speech, language, and hearing research: JSLHR*, vol. 43, no. 3, pp. 796–809, Jun. 2000.
- [5] Y. Maryn, N. Roy, M. De Bodt, P. Van Cauwenberge, and P. Corthals, "Acoustic measurement of overall voice quality: A meta-analysis," *The Journal of the Acoustical Society of America*, vol. 126, no. 5, pp. 2619–2634, Nov. 2009. [Online]. Available: <http://asa.scitation.org/doi/10.1121/1.3224706>
- [6] B. Barsties V Latoszek, Y. Maryn, E. Gerrits, and M. De Bodt, "The Acoustic Breathiness Index (ABI): A Multivariate Acoustic Model for Breathiness," *Journal of Voice: Official Journal of the Voice Foundation*, vol. 31, no. 4, pp. 511.e11–511.e27, Jul. 2017.
- [7] V. Uloza, N. Ulozaitė-Stanienė, T. Petrauskas, and R. Kregždýtė, "Accuracy of Acoustic Voice Quality Index Captured With a Smartphone - Measurements With Added Ambient Noise," *Journal of Voice: Official Journal of the Voice Foundation*, vol. 37, no. 3, pp. 465.e19–465.e26, May 2023.
- [8] Y. Maryn, "Acoustic measurement of overall voice quality in sustained vowels and continuous speech," Ph.D. dissertation, Ghent University, Faculty of Medicine and Health Sciences, Department of Otorhinolaryngology, 01 2010.
- [9] M. R. Schroeder, "New Method of Measuring Reverberation Time," *The Journal of the Acoustical Society of America*, vol. 37, no. 3, pp. 409–412, 03 1965. [Online]. Available: <https://doi.org/10.1121/1.1909343>
- [10] J. Mayer, "VOXplot." [Online]. Available: <https://voxplot.lingphon.com/en>
- [11] B. Woldert-Jokisz, "Saarbrücker Stimmdatenbank," May 2007, publisher: Institut für Phonetik, Universität des Saarlandes. [Online]. Available: https://stimmdatenbank.coli.uni-saarland.de/help_de.php4
- [12] A. F. Zuur, E. N. Ieno, N. Walker, A. A. Saveliev, and G. M. Smith, *Mixed effects models and extensions in ecology with R*, ser. Statistics for Biology and Health. New York, NY: Springer, 2009. [Online]. Available: <http://link.springer.com/10.1007/978-0-387-87458-6>
- [13] M. Kuhn and K. Johnson, *Applied Predictive Modeling*. New York, NY: Springer New York, 2013. [Online]. Available: <http://link.springer.com/10.1007/978-1-4614-6849-3>
- [14] GKV-Spitzenverband, "Anlage 5 Zulassungsvoraussetzungen zum Vertrag nach § 125 Absatz 1 SGB V über die Versorgung mit Leistungen der Stimm-, Sprech-, Sprach- und Schlucktherapie und deren Vergütung vom 15.03.2021," 2021. [Online]. Available: https://www.gkv-heilmittel.de/fuer_heilmittelerbringer/vertraege/vertraege.jsp
- [15] Deutsche Institut für Normung e.V. (Ed), "DIN 18041:2016-03, Hörsamkeit in Räumen.- Anforderungen, Empfehlungen und Hinweise für die Planung," Berlin: Beuth Verlag 2016.
- [16] D. Moore, N. Campbell, S. Rosen, D.-E. Bamiou, T. Sirimanna, P. Grant, and K. Wakeham, "Position statement and practice guidance auditory processing disorder (apd)," 02 2018.